■ Roots

Roots[*lhs*==*rhs*, *var*] yields a disjunction of equations which represent the roots of a polynomial equation.

Roots uses Factor and Decompose in trying to find roots. ■ You can find numerical values of the roots by applying N. ■ Roots can take the following options:

Cubics	True	whether to generate explicit solutions for cubics
EquatedTo	Null	expression to which the variable solved for should be equated
Modulus	Infinity	integer modulus
Multiplicity	1	multiplicity in final list of solutions
Quartics	True	whether to generate explicit solutions for quartics
Using	True	subsidiary equations to be solved

■ Roots is generated when Solve and related functions cannot produce explicit solutions. Options are often given in such cases. ■ Roots gives several identical equations when roots with multiplicity greater than one occur. ■ See page 393.
■ See also: NRoots, Solve, FindRoot, ToRules.

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